

LAB | Deploy a 3-Tier Application on Amazon EKS Cluster

```
seife@SeifEddine:/mnt/c/Users/seife$ kubectl get nodes
NAME                                STATUS    ROLES    AGE   VERSION
ip-10-0-1-207.eu-central-1.compute.internal Ready    <none>   15h   v1.33.12-eks-0de9cde
ip-10-0-2-73.eu-central-1.compute.internal Ready    <none>   15h   v1.33.12-eks-0de9cde
seife@SeifEddine:/mnt/c/Users/seife$ cd ~/projects
seife@SeifEddine:~/projects$ git clone https://github.com/Pokfinner/3-tier-application.git
Cloning into '3-tier-application'...
remote: Enumerating objects: 493, done.
remote: Counting objects: 100% (180/180), done.
remote: Compressing objects: 100% (80/80), done.
remote: Total 493 (delta 130), reused 100 (delta 100), pack-reused 313 (from 2)
Receiving objects: 100% (493/493), 2.15 MiB | 764.00 KiB/s, done.
Resolving deltas: 100% (223/223), done.
seife@SeifEddine:~/projects$ cd 3-tier-application
seife@SeifEddine:~/projects/3-tier-application$ ls
api  docker-compose.yml  running-locally-docker.md  ui
seife@SeifEddine:~/projects/3-tier-application$
```

```
seife@SeifEddine:~/projects/3-tier-application/api$ docker build -t tomioka05/basic3tier -api:latest .
[+] Building 42.5s (5/15)                                docker:default
=> [internal] load build definition from Dockerfile      0.1s
=> => transferring dockerfile: 466B                      0.0s
=> [internal] load metadata for mcr.microsoft.com/dotnet/sdk:7.0 1.2s
=> [internal] load metadata for mcr.microsoft.com/dotnet/aspnet:7.0 1.1s
=> [internal] load .dockerignore                        0.1s
=> => transferring context: 382B                          0.0s
=> [build 1/5] FROM mcr.microsoft.com/dotnet/sdk:7.0@sha256:d32bd65cf5843f413e8 41.1s
=> => resolve mcr.microsoft.com/dotnet/sdk:7.0@sha256:d32bd65cf5843f413e81f5d917 0.1s
=> => sha256:c549dbd140e5a09bcc7e4b1da89ed4b96a0340a730303fbe2 13.99MB / 13.99MB 8.5s
=> => sha256:5e263829ce76bc29ff631384bd5e93356d28625461ca507fa 7.34MB / 180.57MB 8.1s
=> [internal] load build context                        0.2s
=> => transferring context: 61.13kB                      0.1s
=> [base 1/2] FROM mcr.microsoft.com/dotnet/aspnet:7.0@sha256:c7d9ee6cd01afe9aa 41.1s
=> => resolve mcr.microsoft.com/dotnet/aspnet:7.0@sha256:c7d9ee6cd01afe9aa80642e 0.1s
=> => sha256:f194078e85f8008c084163778aaf266434f7182e0d4f78364 8.39MB / 10.12MB 40.6s
=> => sha256:f1b39e168c1c776458e172f157167607b9fd3cc550af8e6ff0a7dd3 153B / 153B 2.9s
=> => sha256:534ba947de6ac79fd6168f4a93847954b23bab2782700bbf 20.97MB / 32.46MB 40.5s
=> => sha256:82bb7a80de578404d92b5ae5e67f3de90eab30027694d260 14.97MB / 14.97MB 23.8s
=> => sha256:728328ac3bde9b85225b1f0d60f5c149f5635a191f5d8eae 14.68MB / 31.43MB 37.6s
```

```

Login Succeeded
seife@SeifEddine:~/projects/3-tier-application/api$ docker push tomioka05/basic3tier-api:latest
The push refers to repository [docker.io/tomioka05/basic3tier-api]
11acbc0b0028: Pushed
534ba947de6a: Pushed
728328ac3bde: Pushed
82bb7a80de57: Pushed
f194078e85f8: Pushed
4f4fb700ef54: Mounted from library/sonarqube
1426f932e870: Pushed
f1b39e168c1c: Pushed
84fd9d0ba580: Pushed
latest: digest: sha256:9b337ad07dd1af90d2a81c2d5deaf2788bbe9b54f35704980c6d729880c773bf
size: 856
seife@SeifEddine:~/projects/3-tier-application/api$

```

Local		My Hub				
tomioka05		Search				
	Tags	OS	Vulnerabilities	Last pushed	Size	Actions
 tomioka05/basic3tier-ui	latest		 Inactive	2 minutes ago	26.66 MB	Pull
 tomioka05/basic3tier-api	latest		 Inactive	16 minutes ago	105.34 MB	Pull
 tomioka05/lab-frontend	1.0		 Inactive	7 days ago	28.52 MB	Pull
 tomioka05/lab-backend	1.0		 Inactive	7 days ago	50.79 MB	Pull
 tomioka05/worker-app	latest		 Inactive	21 days ago	81.26 MB	Pull

```

seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
api-deployment-795f86864b-gg8f9    1/1     Running   0           115s
api-deployment-795f86864b-jqc77    1/1     Running   0           115s
api-deployment-795f86864b-tnvgr     1/1     Running   0           115s
postgres-deployment-fb7b67756-4hg5d 0/1     Pending   0           2m16s
ui-deployment-756cd9878f-xx2hl      1/1     Running   0           112s
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get svc
NAME            TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
api-service     ClusterIP   172.20.208.72 <none>       80/TCP     2m7s
kubernetes      ClusterIP   172.20.0.1    <none>       443/TCP    8d
postgres-service ClusterIP   172.20.1.112  <none>       5432/TCP   2m28s
ui-service      ClusterIP   172.20.114.68 <none>       80/TCP     2m4s
seife@SeifEddine:~/projects/3-tier-application/k8s$

```

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get svc
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)    AGE
api-service         ClusterIP   172.20.208.72 <none>       80/TCP     19m
kubernetes           ClusterIP   172.20.0.1     <none>       443/TCP    8d
postgres-service    ClusterIP   172.20.1.112   <none>       5432/TCP   20m
ui-service           ClusterIP   172.20.114.68  <none>       80/TCP     19m
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get nodes
NAME                                                    STATUS    ROLES    AGE   VERSION
ip-10-0-1-207.eu-central-1.compute.internal            Ready    <none>   18h   v1.33.12-eks-0de9cde
ip-10-0-2-73.eu-central-1.compute.internal             Ready    <none>   18h   v1.33.12-eks-0de9cde
seife@SeifEddine:~/projects/3-tier-application/k8s$
```

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl describe pvc postgres-pvc
Name:                postgres-pvc
Namespace:           default
StorageClass:        gp2
Status:              Pending
Volume:
Labels:              <none>
Annotations:          volume.beta.kubernetes.io/storage-provisioner: ebs.csi.aws.com
                     volume.kubernetes.io/selected-node: ip-10-0-2-73.eu-central-1.compute.int
                     ernal
                     volume.kubernetes.io/storage-provisioner: ebs.csi.aws.com
Finalizers:          [kubernetes.io/pvc-protection]
Capacity:
Access Modes:
VolumeMode:          Filesystem
Used By:              postgres-deployment-fb7b67756-4hg5d
Events:
  Type      Reason              Age           From          Message
  ----      -
  -
  Normal    WaitForFirstConsumer 25m           persistentvolume-controller waiting for first consumer to be created before binding
  Normal    ExternalProvisioning 27s (x102 over 25m) persistentvolume-controller waiting for a volume to be created either by the external provisioner 'ebs.csi.aws.com' or manually by the system administrator. If volume creation is delayed, please verify that the provisioner is running and correctly registered.
seife@SeifEddine:~/projects/3-tier-application/k8s$
```

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get pods -n kube-system
```

NAME	READY	STATUS	RESTARTS	AGE
aws-node-p828k	2/2	Running	0	19h
aws-node-v52xc	2/2	Running	0	19h
coredns-84fd587d5-8znbl	1/1	Running	0	19h
coredns-84fd587d5-98vsh	1/1	Running	0	19h
ebs-csi-controller-7b8c59447-9lmf8	6/6	Running	0	3m6s
ebs-csi-controller-7b8c59447-gb4dk	6/6	Running	0	3m6s
ebs-csi-node-cjjsz	3/3	Running	0	43m
ebs-csi-node-pvwj2	3/3	Running	0	43m
kube-proxy-ffftz	1/1	Running	0	19h
kube-proxy-m7glb	1/1	Running	0	19h

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get pods pvc
Error from server (NotFound): pods "pvc" not found
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get pvc
```

NAME	STATUS	VOLUME	CAPACITY	ACCESS MOD
ES	STORAGECLASS	VOLUMEATTRIBUTESCLASS	AGE	
postgres-pvc	Bound	pvc-6230a5c8-2425-4d47-95ea-18840d5ffffd	5Gi	RWO
gp2	<unset>	88m		

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
api-deployment-795f86864b-gg8f9	1/1	Running	0	117m
api-deployment-795f86864b-jqc77	1/1	Running	0	117m
api-deployment-795f86864b-tnvgr	1/1	Running	0	117m
postgres-deployment-645b79df64-rh452	1/1	Running	0	17m
ui-deployment-756cd9878f-xx2hl	1/1	Running	0	117m

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get nodes
```

NAME	STATUS	ROLES	AGE	VERSION
ip-10-0-1-207.eu-central-1.compute.internal	Ready	<none>	20h	v1.33.12-eks-0de9cde
ip-10-0-2-73.eu-central-1.compute.internal	Ready	<none>	20h	v1.33.12-eks-0de9cde

```
seife@SeifEddine:~/projects/3-tier-application/k8s$ aws eks update-kubeconfig --region eu-central-1 --name eks-seif
Updated context arn:aws:eks:eu-central-1:686699774218:cluster/eks-seif in /home/seife/.kube/config
seife@SeifEddine:~/projects/3-tier-application/k8s$ kubectl get ingress
```

NAME	CLASS	HOSTS	ADDRESS	PORTS	AGE
ui-ingress	nginx	*	a68426cd3b1b641f29ccc3cf766238b8-4be93a8ea105ae6d.elb.eu-central-1.amazonaws.com	80	4m38s

```
seife@SeifEddine:~/projects/3-tier-application/k8s$
```

Basic 3 Tier App

Add

ID	Name	Email	Age	Address	Actions
----	------	-------	-----	---------	---------